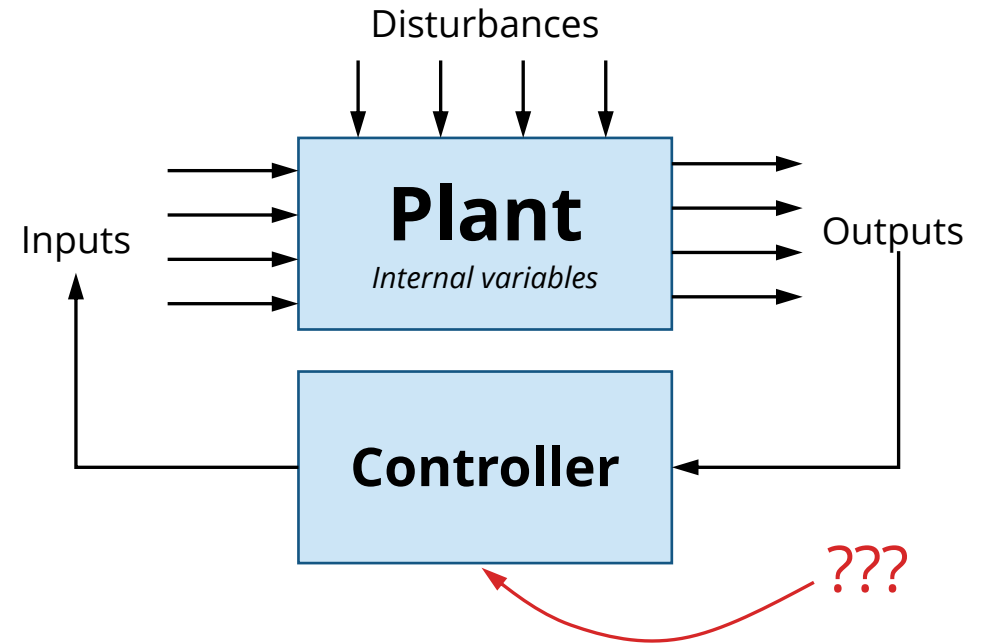




AA 513 Multivariable Control

Lecture 09 // Week 9

Instructor: Prof. Karen Leung

Announcements

- Research paper. **Sign up for a slot!**
- Tutorial. Due Thursday **Dec 11 midnight**
- Tutorial peer review. Due **Dec 15 noon**
 - Tutorial to review will be assigned later.
- Exercise 13 & 14 out
 - Due **Monday Dec 1 midnight**

Course evaluations available now

Please take a few minutes to do the course evaluation. Should have received an email with link

Some context

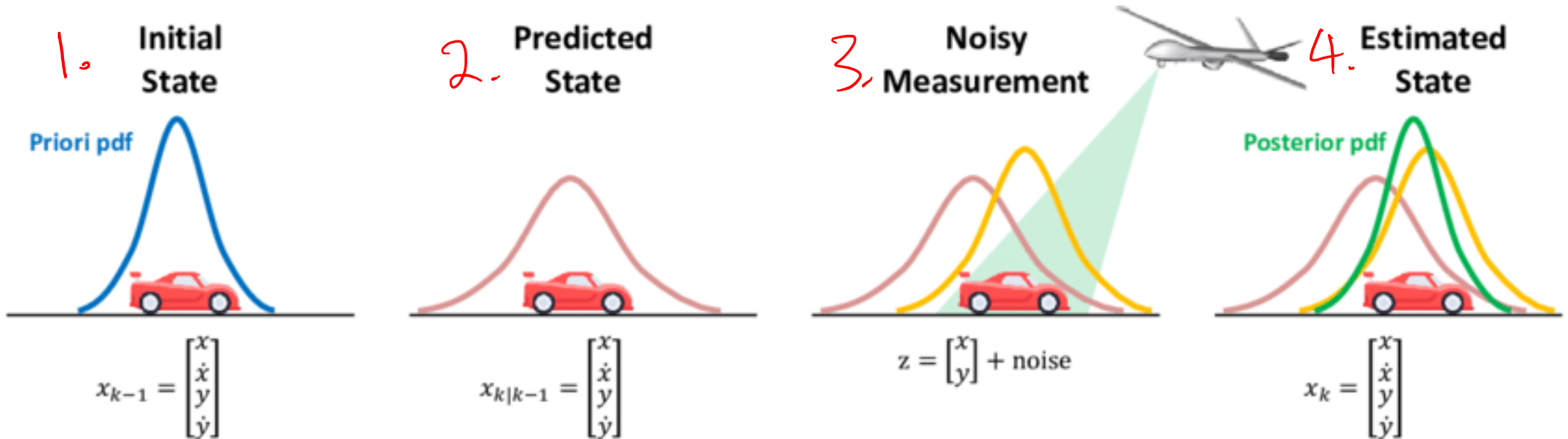
- First time I am teaching course in the evening / mostly remote
- Typically I write on the whiteboard
- Open to feedback regarding slides and online format
- Significant update on exercises

Last time

- **Trajectory optimization:** techniques for solving an optimal trajectory (control/state sequence/signal) over a horizon, subject to constraints
 - Differs from LQR (how?)
- **Transcription methods:** techniques for turning continuous-time problem (infinite dimensional problem) into a discrete one (finite dimensional problem)
 - Discretize controls (and states) to create “nodes”. Reflects a discrete-time formulation
 - Apply simple integration schemes on dynamics to get algebraic representations
 - Approximate control and state signals as polynomials & optimize over coefficients
- Boils down to a **nonlinear program (NLP)**. Still need to solve it
 - One approach: **Sequential Quadratic Program**
 - Repeatedly approximate as QP and solve.
- Many other ways to solve for a trajectory
 - **MPPI:** sampling-based approach

This week

- State estimation!
 - Techniques to estimate the true state of the system given observations about it.

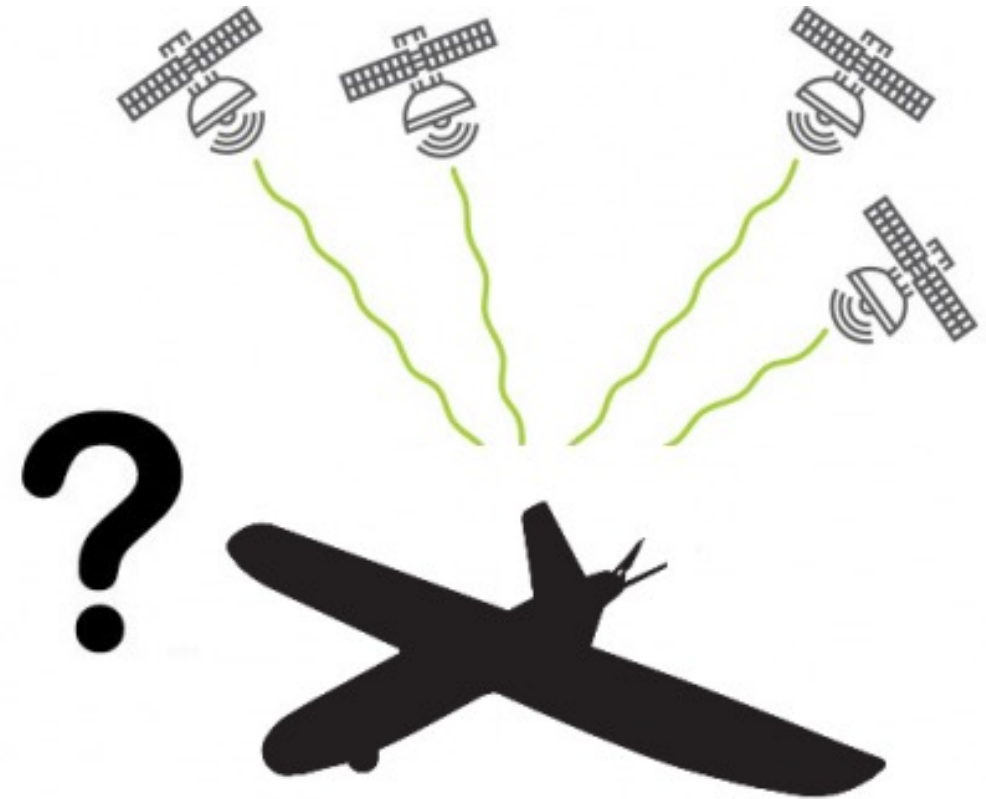


So far in this course...

- We have been assuming that we knew the state perfectly!
 - E.g., position, velocity, heading, etc. $\vec{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \rightarrow \text{LQR on } \vec{x}$
- But that's not always true! Not all states are directly observable!

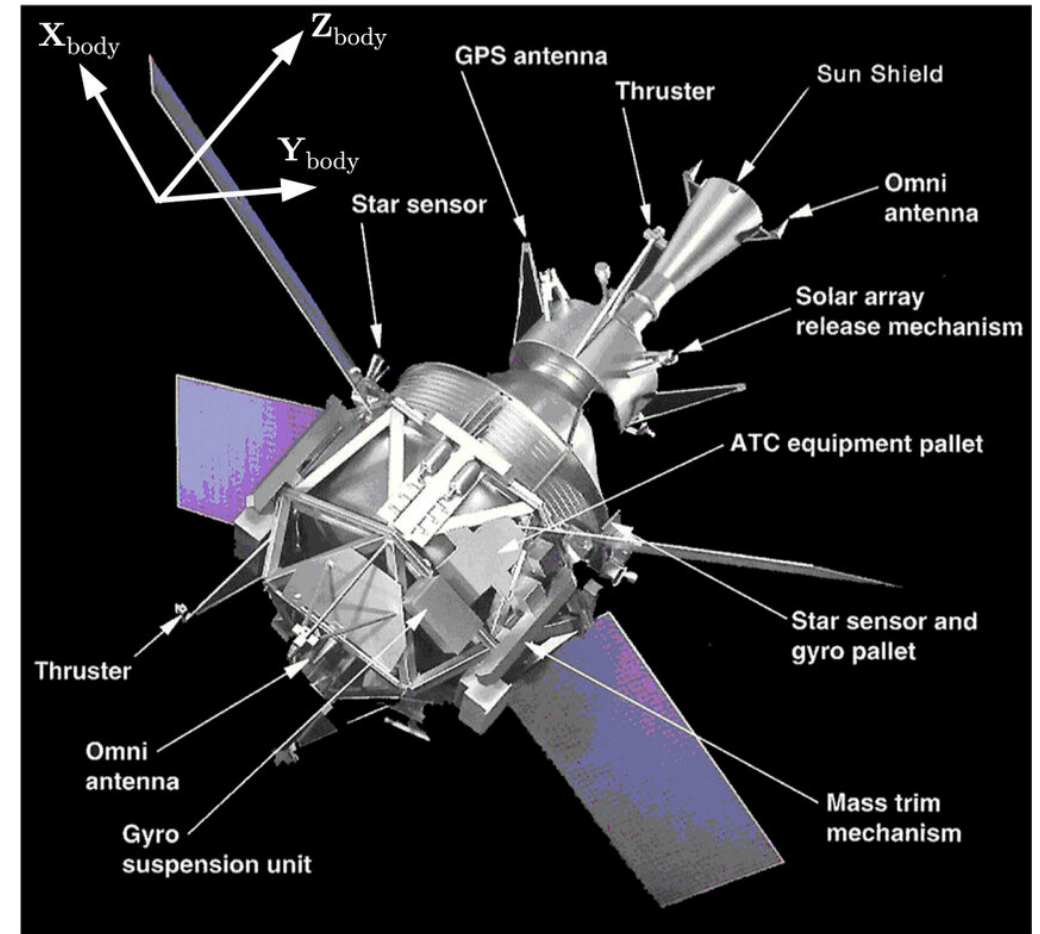
E.g., Aircraft attitude and navigation

- No sensor directly measures attitude — gyros drift, accelerometers are noisy, magnetometers are distorted.
- GPS provides delayed, low-rate, or occasionally unavailable position updates.



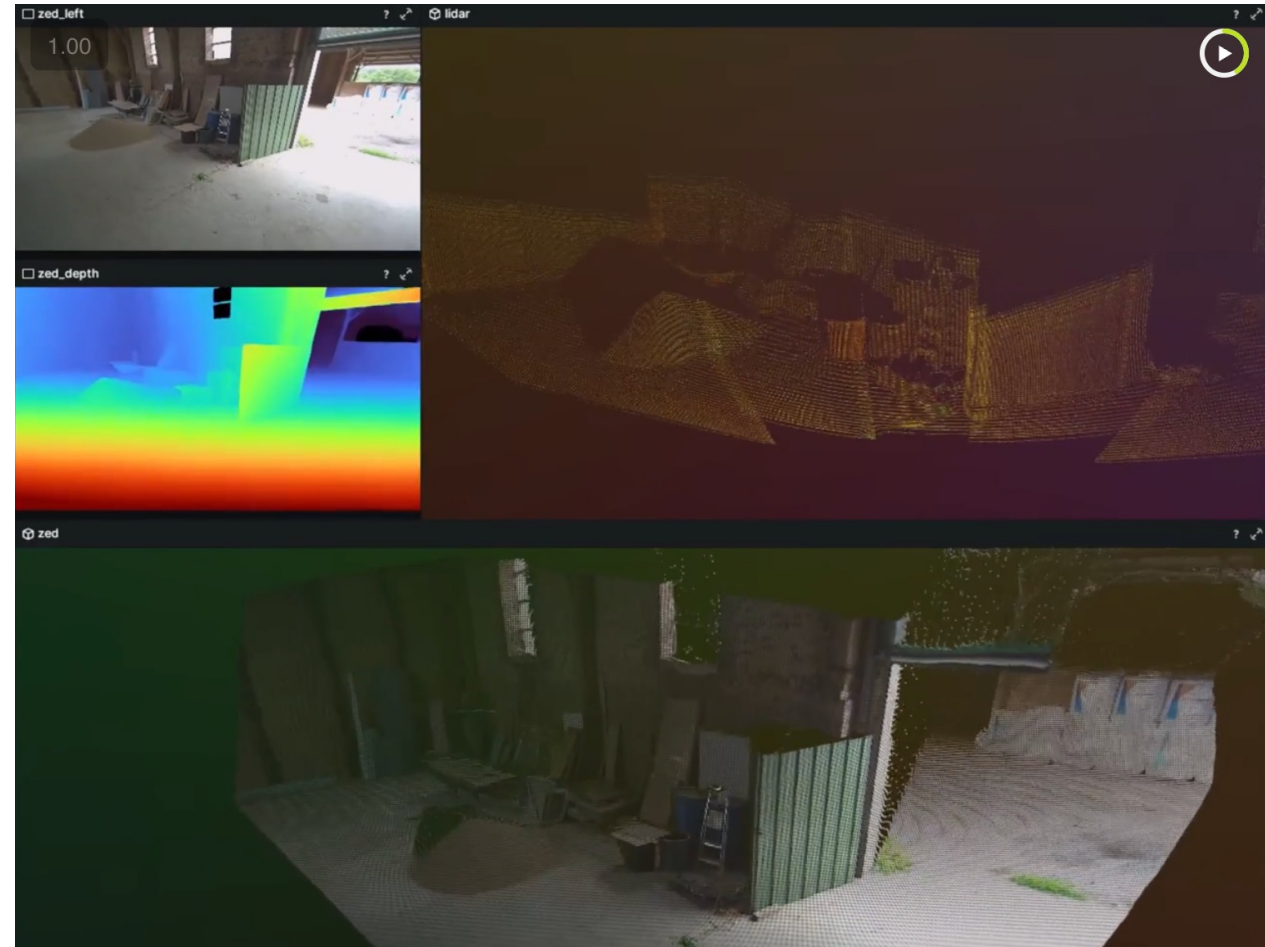
Spacecraft orbit determination

- **States not directly observable:** spacecraft 3D position and velocity relative to Earth or another body.
- Ground stations only measure *range, range-rate, or angles* (azimuth/elevation). These measurements are sparse, noisy, and cannot uniquely determine the full 6-DOF state.
- IMUs, gyroscopes, star trackers, sun sensors, and magnetometers help estimate attitude



Robotics

- Need to know exact position and orientation in the environment
- Odometry is not accurate, especially over time
- Environment is not assumed to be known beforehand
- Common to estimate both robot state AND environment simultaneously (SLAM)
- Common to use cameras, LiDAR, and radar

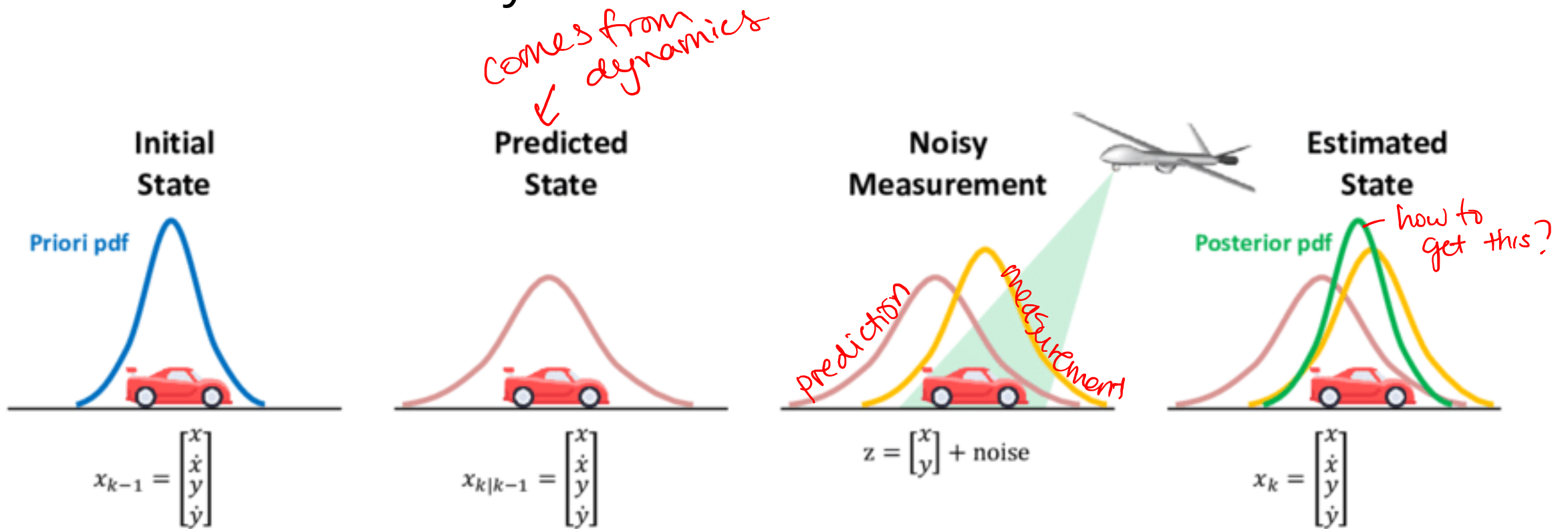


General intuition behind state estimation techniques: requirements

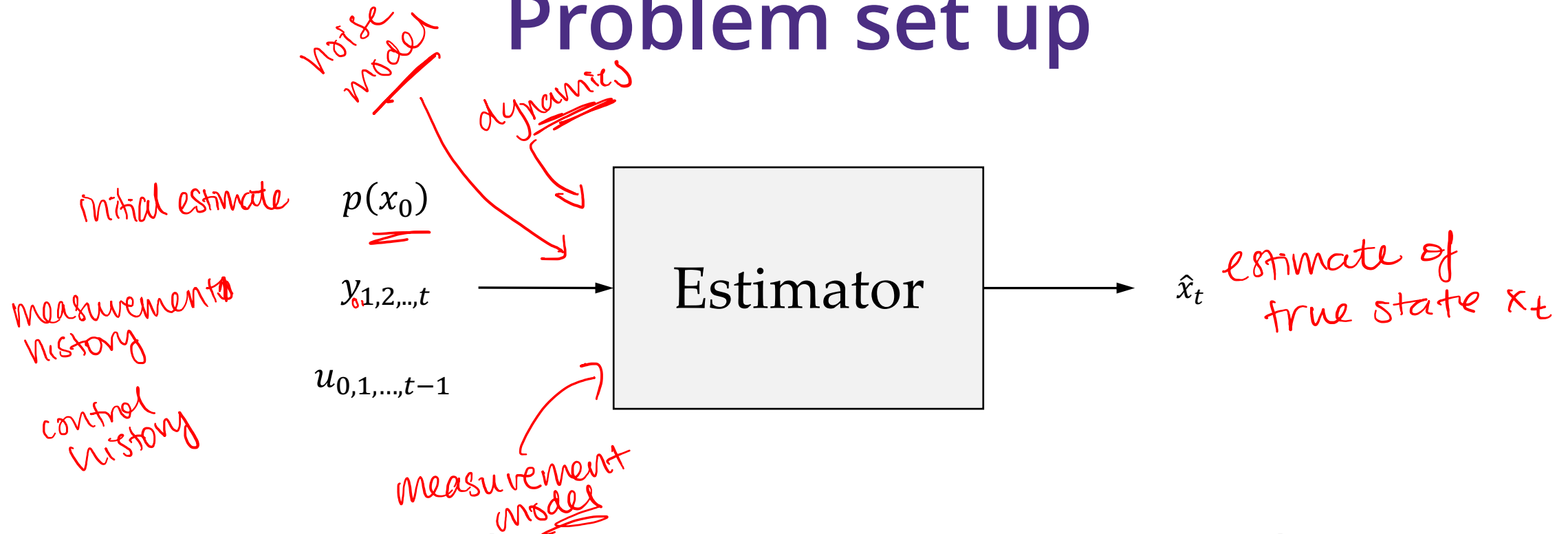
- **Dynamics model**: Something to describe how the state evolves over time. This typically includes robot dynamics, but potentially also environment. $\dot{x} = f(x, u, \omega)$
- **Measurement model**: How do the measurements offer insight into the state
- **Noise/uncertainty model**: The sensors are not perfect. Dynamics are not perfect. Need some model for describing this imperfection.
- **Initial estimation**: Need some sort of initial guess to begin with (prior)

General intuition behind state estimation techniques: Approach

- Consolidate both dynamics and measurement information



Problem set up



Want to make sure $|x_t - \hat{x}_t|$ is small as possible.

(discrete-time)

Kalman Filter: Problem set up

- Gaussian linear dynamics

system dyn.

$$x_{t+1} = Ax_t + Bu_t + \boxed{w_t}, \text{ additive noise}$$

measurement model

$$y_t = Cx_t + \boxed{v_t}, \text{ additive noise}$$

White noise,

$w_t \perp w_{t+1}$

Covariance

$$\mathbb{E}[w_t w_t^T] = Q, \quad w_t \sim \mathcal{N}(0, Q) \text{ zero mean.}$$

$$\mathbb{E}[v_t v_t^T] = R, \quad v_t \sim \mathcal{N}(0, R) \text{ zero mean.}$$

- Estimate of state

state estimate at t

$$\hat{x}_t \sim p(\hat{x}_t \mid \overset{\text{past control}}{u_{0:t-1}}, \overset{\text{past meas.}}{y_{1:t}}, \overset{\text{initial state}}{x_0})$$

estimate is Gaussian.

$$\hat{x}_t \sim \mathcal{N}(\mu_t, P_t)$$

- Covariance of error $P_t = \mathbb{E}[e_t e_t^T] = \mathbb{E}[\underbrace{(x_t - \hat{x}_t)}_{e_t} (x_t - \hat{x}_t)^T]$

- What is μ_t, P_t ? Depend on previous values.

$$\mu_t = g(u_{t-1}, P_{t-1}, u_{t-1}, y_{t-1}, Q, R), \quad P_t = h(\mu_{t-1}, P_{t-1}, \dots)$$

Restating the notation

- True state and dynamics: & measurement

true state x_t

→ true dynamics. $x_{t+1} = Ax_t + Bu_t + \underline{w_t}$, $\text{Cov}(w_t) = Q$

→ true measurement $y_t = Cx_t + v_t$, $\text{Cov}(v_t) = R$

- State estimate:

state estimate: $\hat{x}_t$, estimate of x . $\hat{x}_t$ is a random variable.

$\hat{x}_t \sim \mathcal{N}(\mu_t, P_t)$, μ_t is mean, P_t is covariance

- Estimation error:

$e_t = x_t - \hat{x}_t$, e_t is also a random variable

$$\boxed{\text{Cov}(e_t) = P_t} \quad \mathbb{E}[e_t e_t^T] = P_t$$

Step 0: Initial estimate $\hat{x}_0 \sim P(x_0)$

no conditional
argument

the "expert" sets an initial distribution

$$\hat{x}_0 \sim \mathcal{N}(\mu_0, \rho_0)$$

You need to provide values
for these.

Step 1: Predict estimate at next time step

Operation: Estimate state estimate at next time step using dynamics only

- passing a Gaussian through a linear transformation will output another Gaussian w/ different mean + covariance.

predicted mean: $\mu_{t-1} \rightarrow \boxed{\text{dynamics}} \rightarrow \mu_t = \dots ?$

$\mu_t^P = A \mu_{t-1} + B u_{t-1}$

$(\mu_{t+1}^P = A \mu_t + B u_t)$

what is the predicted covariance? P_t^p

$\text{cov}(e_t^p)$ = P_t^p , $e_t^p = \underline{x}_t - \hat{x}_t^p$
 expand.

$$x_t = Ax_{t-1} + Bu_{t-1} + w_{t-1}$$

$$\hat{x}_t^p = \hat{A} \hat{x}_{t-1} + Bu_{t-1} \leftarrow \text{predict estimate}$$

no p superscript

$$\mathbb{E}[e_t^p e_t^{pT}] = \mathbb{E}[(x_t - \hat{x}_t^p)(x_t - \hat{x}_t^p)^T]$$

$$= \mathbb{E}[(\underline{A}x_{t-1} + \cancel{Bu_{t-1}} + w_{t-1} - \hat{A}\hat{x}_{t-1} - \cancel{Bu_{t-1}})(\dots)^T]$$

$$= \mathbb{E}[(\underline{A}(\underline{x}_{t-1} - \hat{x}_{t-1}) + w_{t-1})(\underline{A}(\underline{x}_{t-1} - \hat{x}_{t-1}) + w_{t-1})^T]$$

e_{t-1}

$$e_{t-1} = x_{t-1} - \hat{x}_{t-1}$$

$$= \mathbb{E}[(\underline{A}e_{t-1} + w_{t-1})(\underline{A}e_{t-1} + w_{t-1})^T] = \mathbb{E}[\underbrace{\underline{A}e_{t-1}e_{t-1}^T \underline{A}^T}_{P_{t-1}} + \underbrace{w_{t-1}w_{t-1}^T}_{Q_{t-1} = Q}]$$

$w_{t-1} \perp e_{t-1}$

P_{t-1}

$Q_{t-1} = Q$

$$P_t^p = A P_{t-1} A^T + Q \rightarrow \text{predicted covariance at next timestep.}$$

Step 2: Receive measurement

Operation: Compute measurement residual (or *innovation vector*). Error between measurement and *predicted* measurement.

y_t is the measurement at t .

$C\hat{x}_t^P$ is the predicted measurement at t

$$h_t = y_t - C\hat{x}_t^P \rightarrow \text{measurement residual.}$$

Step 3: Update estimate given measurement

Operation: Update state error at next time step given measurement information

$$\hat{\mathbf{x}}_t = \hat{\mathbf{x}}_t^p + K_t (y_t - C \hat{\mathbf{x}}_t^p)$$

Annotations for the equation above:

- $\hat{\mathbf{x}}_t$: best estimate of state at t
- $\hat{\mathbf{x}}_t^p$: predicted estimate (step 1)
- K_t : Kalman Gain trade-off measurement & dyn. uncertainty
- $(y_t - C \hat{\mathbf{x}}_t^p)$: step 2 measurement residual.
- $\hat{\mathbf{x}}_t \sim \mathcal{N}(\mu_t, P_t)$

updated ~~new~~ mean of estimate:

$$\mu_t = \mu_t^p + K_t (y_t - C \mu_t^p)$$

updated covariance: P_t | ~~not~~ true measurement $y_t =$

$$E[e_t e_t^T] = P_t$$

$$E[e_t e_t^T] = E[(x_t - \hat{x}_t)(x_t - \hat{x}_t)^T]$$

$$= E[(x_t - \hat{x}_t^P - K_t(y_t - C\hat{x}_t^P))(\dots)^T]$$

$$= E[(\underbrace{x_t - \hat{x}_t^P}_{\dots} - K_t(\underbrace{C x_t + v_t}_{\dots} - \underbrace{C \hat{x}_t^P}_{\dots}))(\dots)^T]$$

$$= E[\underbrace{((I - K_t C)(x_t - \hat{x}_t^P) - K_t v_t)}_{\dots}(\dots)^T] \quad (\text{cross terms cancel})$$

$$= E[(I - K_t C)(x_t - \hat{x}_t^P)(x_t - \hat{x}_t^P)^T(I - K_t C)^T + K_t \underbrace{v_t v_t^T}_{R} K_t^T]$$

$\underbrace{(x_t - \hat{x}_t^P)(x_t - \hat{x}_t^P)^T}_{e_t^P e_t^{PT}}$

$$P_t = (I - K_t C) P_t^P (I - K_t C)^T + K_t R K_t^T \quad \leftarrow \text{plug } K_t^* \text{ here.}$$

what is K_t ?

want to pick K_t such that covariance is as "small" as possible.

$$P_t = \begin{bmatrix} e_{t1} e_{t1}^T & & \\ & e_{t2} e_{t2}^T & \\ \text{cross terms} & & \ddots & \\ & & & e_{tn} e_{tn}^T \end{bmatrix} \Rightarrow \text{want } e_{ti} e_{ti}^T \text{ (ie. variance of each variable) to be as small as possible.}$$

mostly care about the size of the diagonal entries.

~~sum~~ sum of diagonal entries. $\Rightarrow$ TRACE!

$$\min_{K_t} \text{Tr}(P_t)$$

.....

$$K_t^* = P_t^P C^T (C P_t^P C^T + R)^{-1}$$

optimal gain.
aka Kalman gain.

$$\Rightarrow P_t = (I - K_t^* C) P_t^P$$

Kalman filter steps: Initialize

- Initialize state estimate: mean and covariance

$$\mu_0, P_0$$

Kalman filter steps: Predict

Predict next state estimate using noise-free dynamics only

$$\mu_t^p = A\mu_{t-1} + Bu_{t-1} \quad P_t^p = AP_{t-1}A^T + Q$$

Kalman filter steps: Receive measurement

Receive measurement & compare with predicted measurement

$$h_t = y_t - C\hat{x}_t^p$$

Kalman filter steps: Update step

- Mean update $\mu_t = \mu_t^p + K_t(y_t - C\mu_t^p)$
- Covariance update $P_t = (I - K_t^* C)P_t^p(I - K_t^* C)^T + K_t^* R K_t^{*T}$
 $K_t^* = P_t^p C^T (C P_t^p C^T + R)^{-1}$
 $P_t = (I - K_t^* C)P_t^p$

any dependence on controls?

no u_t in the equations.

Kalman Filter algorithm

0. Start with initial state estimate $\hat{x}_0 \sim \mathcal{N}(\mu_0, P_0)$

For $t = 1, 2, \dots$

- 
1. Predict step $\mu_{t-1}, P_{t-1} \rightarrow \mu_t^p, P_t^p$
~ get measurement
 2. Update step $\mu_t^p, P_t^p \rightarrow \mu_t, P_t$

- Iterate ***forward*** in time

Alternate derivation (sketch)

- We can also derive the same set of equations via a *Bayesian perspective*

$$\boxed{p(x | y)} = \frac{\boxed{p(y | x)} \boxed{p(x)}}{p(y)}$$

state / meas. (above $p(x | y)$)
 measurement (above $p(y | x)$)
 state (above $p(x)$)
 prior. (above $p(x)$)
 Bayes Rule. (to the right)
 posterior (below $p(x | y)$)
 measurement model. (below $p(y | x)$)

- Kalman filter is a recursive Bayesian filter, with linear models and Gaussian noise

$$\rightarrow p(\underline{x_t} | \underline{y_{0:t}}) \sim \underline{p(y_t | x_t)} \underline{p(x_t | y_{0:t-1})}$$

$u_{0:t}$ (above $y_{0:t}$)
 u_t (above x_t)
 $u_{0:t-1}$ (above $y_{0:t-1}$)

due to Markovian nature of our system

next state only depends on previous state & control.

Connections with LQR

- (Aside from the deliberate notation choice of Q , R , K)
- **LQR controller**
 - Runs backward in time
 - Has Riccati equations.
- **Kalman filter**
 - Runs forward in time
 - Riccati equation? 🤔

Is the Kalman filter secretly a Riccati equation?

$$P_t = (I - K_t^* C) P_t^p$$

$$P_{t-1} = (I - K_{t-1}^* C) P_{t-1}^p$$

predict step.

$$P_t^p = A P_{t-1} A^T + Q$$

we get this

$$P_{t+1}^p = Q + A P_t^p A^T - A P_t^p C^T (C P_t^p C^T + R)^{-1} C P_t^p A^T$$

forward in time

discrete time LQR

From LQR

backward

$$P_{t-1} = Q + A^T P_t A - A^T P_t B (B^T P_t B + R)^{-1} B^T P_t A$$

$$K_t = (B^T P_{t+1} B + R)^{-1} B^T P_{t+1} A$$

Is the Kalman filter secretly a Riccati equation?

YES!

Duality! There is a connection

LQR Description	LQR	Kalman filter	KF Description
Dynamics	A	A^T	Dynamics transposed
Control matrix	B	C^T	Observation transposed
State cost	Q	Q	Process noise covariance
Control cost	R	R	Measurement noise covariance
Terminal cost	Q_T	P_0	Initial state covariance
Value function	P_t	P_t	Error covariance

Continuous time Kalman Filter

- Continuous time dynamics

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t), \quad y(t) = Cx(t) + v(t)$$

$$w(t) \sim N(0, Q(t)), v(t) \sim N(0, R(t))$$

- Turn into discrete-time dynamics

$$x_{t+1} = (I + A\Delta t)x_t + B\Delta t u_t + \tilde{w}_t, \quad y_t = Cx_t + \tilde{v}_t$$

$$\tilde{w}_t \sim N(0, Q\Delta t), \quad \tilde{v}_t \sim N(0, \frac{R}{\Delta t})$$

Continuous time Kalman Filter

- (Skipping the algebra.) Some key takeaways

$$\frac{1}{\Delta t} K_t = P_t^p C^T (C P_t^p C \Delta t + R)^{-1} \Rightarrow K_t(t) = P_t^p C^T R^{-1}$$

Kalman gain for CT

- In the derivation, if $\Delta t \rightarrow 0$, then $K_t \rightarrow 0$, but

$$\lim_{\Delta t \rightarrow 0} \frac{K_t}{\Delta t} = P_t^p C^T R^{-1}.$$

- Don't just use discrete time KF for very small timesteps
- Predict & update step occurs simultaneously, i.e., $P^p = P$

$$\lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} (P_t^p - P_{t-1}^p) \Rightarrow \dot{P} = AP + PA^T - PC^T R^{-1} CP + Q$$

Can we apply LQR and KF together?

- Potential risk: Coupling between the two.
-  • Does KF equations depend on control input?
-  • Does LQR depend on covariance?

Separation principle

For a **linear system** with **quadratic cost** and **Gaussian noise**, the problems of **state estimation** and **control** can be designed *independently* and then combined to produce the **optimal** controller.

Linear Quadratic Gaussian (LQG) = LQR + KF

LQG controller: Apply both LQR and KF

- Use KF for optimal state estimation and LQR for optimal control
get μ_t, P_t .
 - For LQR, use mean of estimate *use μ_t for control* $u_t^* = K_{LQR}(\mu_t)$ *usually x_t*
- **Robustness guarantees?** Will the system stabilize if there is model uncertainties or parameter variations (i.e., model mismatch)?

and

$$R=1.$$

Note that the estimation and control problems have identical (dual) solution matrices.

It can be shown analytically that the optimal gain vector g in $u = -g'x$ may be written as a function of q as

$$g = \begin{bmatrix} 1 \\ 1 \end{bmatrix} (2 + \sqrt{4+q}).$$

A similar relation holds between the optimal filter gain k and σ . For simplicity, let

$$g = f \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$k = d \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

where $f \approx 2 + \sqrt{4+q}$ and $d \approx 2 + \sqrt{4+\sigma}$.

Suppose that the resulting closed-loop controller has a scalar gain m (nominally unity) associated with the input matrix. Only the nominal value of this gain is known to the filter. The full system matrix then becomes

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -mf & -mf \\ d & 0 & 1-d & 1 \\ d & 0 & -d-f & 1-f \end{bmatrix}.$$

Evaluation of the characteristic polynomial is rather tedious, but reveals that only the last two terms are functions of m . The linear term is

$$d + f - 4 + 2(m-1)df$$

and the constant term is

$$1 + (1-m)df.$$

A necessary condition for stability is that both terms be positive. It is easy to see that for sufficiently large d and f (or q and σ), the system is unstable for arbitrarily small perturbations in m in either direction. Thus, by choice of q and σ the gain margins may be made arbitrarily small.

It is interesting to note that the margins deteriorate as control weight gets small and/or system driving noise gets large. In modern control folklore, these have often been considered ad hoc means of improving sensitivity.

It is also important to recognize that vanishing margins are not only associated with open-loop unstable systems. It is easy to construct minimum phase, open-loop stable counterexamples for which the margins are arbitrarily small.

The point of these examples is that LQG solutions, unlike LQ solutions, provide no global system-independent guaranteed robustness properties. Like their more classical colleagues, modern LQG designers are obliged to test their margins for each specific design.

It may, however, be possible to improve the robustness of a given design by relaxing the optimality of the filter with respect to error properties. A promising approach appears to be the introduction of certain fictitious system noises in the filter design procedure. This approach will be the topic of future papers.

REFERENCES

- [1] M. G. Safonov and M. Athans, "Gain and phase margins for multiloop LQG regulators," *IEEE Trans. Automat. Contr.*, Apr. 1977.

Guaranteed Margins for LQG Regulators

JOHN C. DOYLE

Abstract—There are none.

INTRODUCTION

Considerable attention has been given lately to the issue of robustness of linear-quadratic (LQ) regulators. The recent work by Safonov and Athans [1] has extended to the multivariable case the now well-known guarantee of 60° phase and 6 dB gain margin for such controllers. However, for even the single-input, single-output case there has remained the question of whether there exist any guaranteed margins for the full LQG (Kalman filter in the loop) regulator. By counterexample, this note answers that question; there are none.

A standard two-state single-input single-output LQG control problem is posed for which the resulting closed-loop regulator has arbitrarily small gain margin.

EXAMPLE

Consider the following:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 1 \end{bmatrix} w$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

Guaranteed Margins for LQG Regulators

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EXAMPLE

Consider the following:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 1 \end{bmatrix} w$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

where (x_1, x_2) , u , and y denote the usual states, control input, and measured output, and where w and v are Gaussian white noises with intensities $\sigma > 0$ and 1, respectively.

Let performance integral have weights

$$Q = qCC^T = q \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad q > 0$$

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Next week: Research paper presentations!

Sign up for a slot if you have not already!